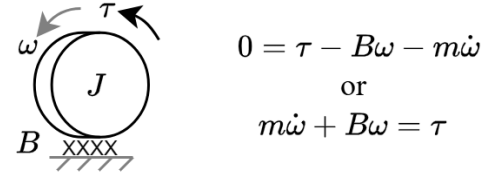
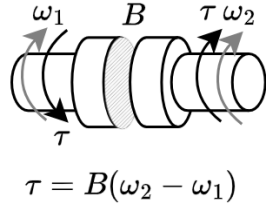
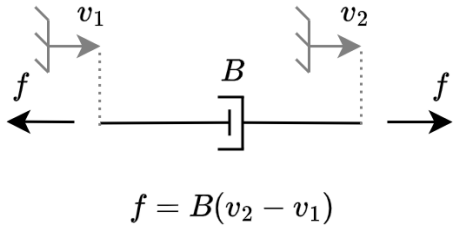
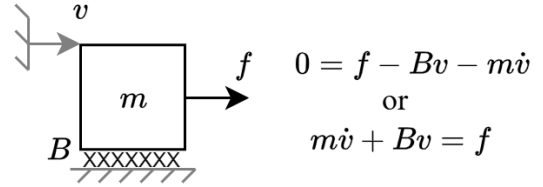
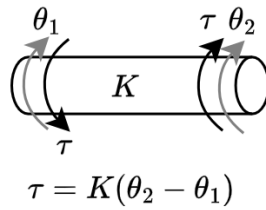
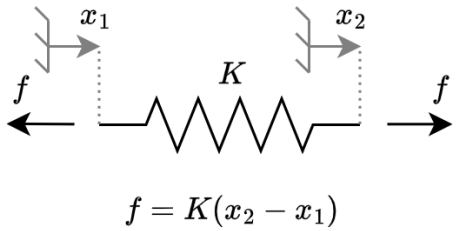
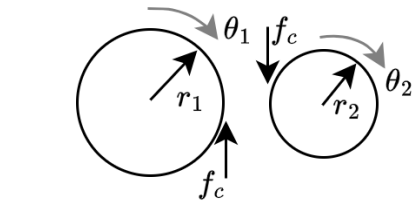
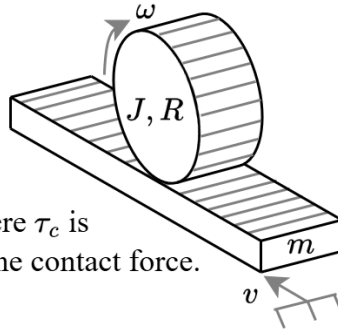


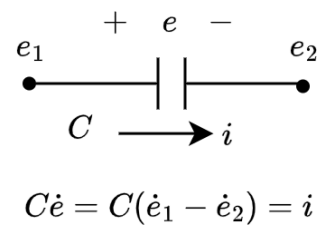
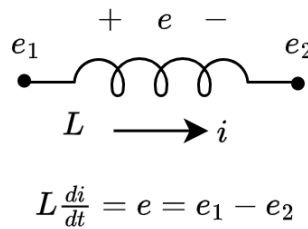
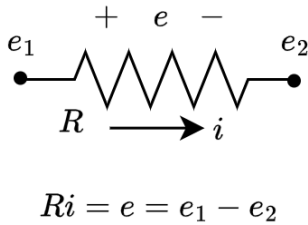
Mechanical Systems:

At the point of contact between the rack and pinion:
 $v = \omega R$ and $\tau_c = -Rf_c$, where τ_c is the contact torque, and f_c is the contact force.

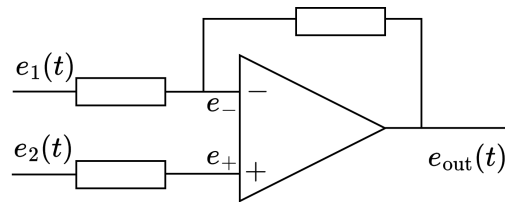


The contact torques are $\tau_1 = r_1 f_c$ and $\tau_2 = r_2 f_c$, and the gear ratio is

$$N = \frac{r_2}{r_1} = -\frac{\omega_1}{\omega_2}$$

Electrical Circuits:Ideal Operational Amplifier Rules:

1. No current flow at the + or - terminals.
2. When the output terminal has a feedback connection to the - terminal, $e_+ = e_-$.

State Space Form:

For a system with n states, m inputs, and p outputs, the state-space form is
$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \\ \mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \end{cases}$$

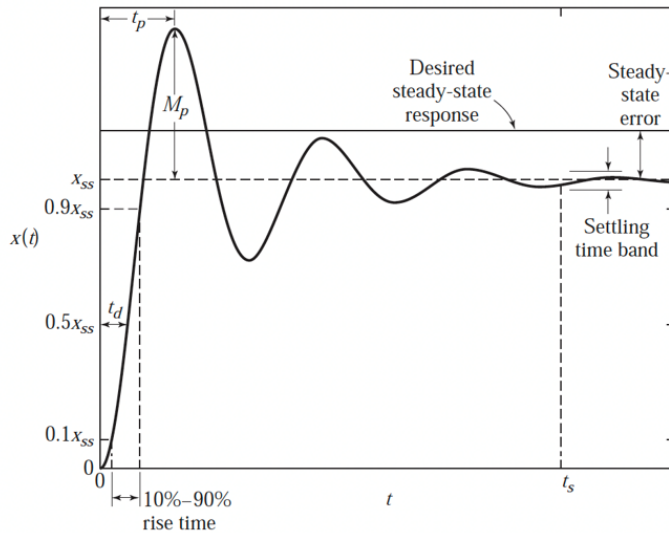
where $\mathbf{A}$ is $n \times n$, $\mathbf{B}$ is $n \times m$, $\mathbf{C}$ is $p \times n$, and $\mathbf{D}$ is $p \times m$.

Linearization:

The incremental variables $\hat{x}(t) = x(t) - \bar{x}$ and $\hat{f}(t) = f(t) - \bar{f}$ linearize a system at the operating point $(\bar{x}, \bar{f})$. A nonlinear function $y(t) = f(x(t))$ may be linearized as $y(t) \approx f(\bar{x}) + \left. \frac{\partial f}{\partial x} \right|_{\bar{x}} \hat{x}(t)$.

First-Order and Second-Order Systems:

- First-order ODE standard form: $\dot{y}(t) + ay(t) = Ku(t)$, where $\tau = 1/a$ is the time constant.
- Second-order ODE standard form: $\ddot{x}(t) + 2\zeta\omega_n\dot{x}(t) + \omega_n^2x(t) = K\omega_n^2u(t)$, where ζ is the damping ratio, and $\omega_n \geq 0$ is the natural frequency.
- For a bounded time-domain function $g(t)$, with Laplace transform $G(s) = \mathcal{L}\{g(t)\}$:
 - Final value theorem: $g(\infty) = \lim_{t \rightarrow \infty} g(t) = \lim_{s \rightarrow 0} sG(s)$
 - Initial value theorem: $g(0^+) = \lim_{t \rightarrow 0^+} g(t) = \lim_{s \rightarrow \infty} sG(s)$
- Settling time: $T_s = 5\tau = -5/\sigma$, where σ is the real part of the pole that is the closest to the $j\omega$ -axis.
- With zero initial-conditions, when a step-input is applied, a generic second-order step-response is



where:

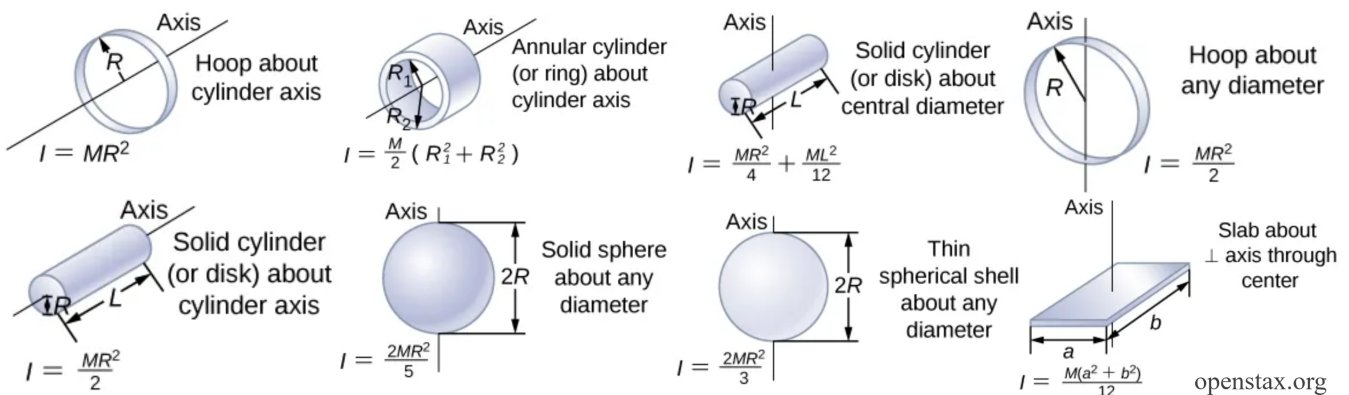
$$M_p = e^{\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right)}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} = \frac{\pi}{\omega_d}$$

$$\%O.S. = e^{\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}\right)} \times 100\%$$

Partial Fraction Expansion:In each applicable case, a , A , B , and C are real-valued numbers.

- Distinct and real roots: *e.g.*, $F(s) = \frac{1}{(s + \sigma_1)(s + \sigma_2)} = \frac{A}{s + \sigma_1} + \frac{B}{s + \sigma_2}$
- Repeated real roots: *e.g.*, $F(s) = \frac{1}{(s + \sigma_1)^2} = \frac{A}{(s + \sigma_1)^2} + \frac{B}{s + \sigma_1}$
- Complex roots: *e.g.*, $F(s) = \frac{s + z}{(s + \sigma - j\omega)(s + \sigma + j\omega)} = \frac{s + z}{s^2 + 2\sigma s + \omega^2} = \frac{Bs + C}{(s + a)^2 + \omega^2}$

Common Moments of Inertia:

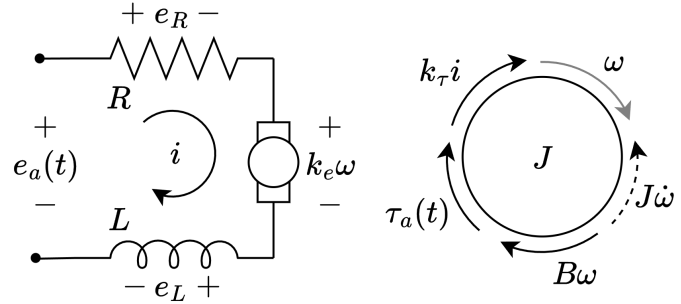
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Permanent Magnet Brushed DC Motors:Electrical Portion:

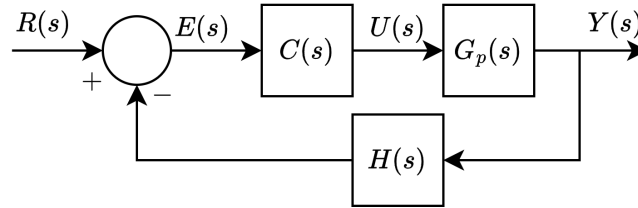
Applied voltage $e_a(t)$, resistance R , inductance L , back-emf constant $k_e = k_\tau$

Mechanical Portion:

Applied torque $\tau_a(t)$, rotational inertia J , damping B , torque constant $k_\tau = k_e$

**Controlled Systems:**

For the block diagram shown, it has output $Y(s)$, reference $R(s)$, controller transfer function $C(s)$, uncontrolled physical plant transfer function $G_p(s)$, and sensor transfer function $H(s)$.



- Its closed-loop transfer function is $G_{CL}(s) = \frac{Y(s)}{R(s)} = \frac{C(s)G_p(s)}{1 + H(s)C(s)G_p(s)}$.
- Its open-loop transfer function is $KG(s) = H(s)C(s)G_p(s)$, where K is the DC-gain of the system.
- Its closed-loop characteristic equation is $0 = 1 + H(s)C(s)G_p(s)$. Thus, $KG(s) = -1$.

For PID control: $C(s) = K_p + K_i \frac{1}{s} + sK_d$ and $u(t) = K_p e(t) + K_i \int e(t)dt + K_d \frac{d}{dt}e(t)$.

Root Locus:

The governing equation is $1 + K^*L(s) = 0$.

- Root locus begins at the (open-loop) poles, and ends at the (open-loop) zeroes.
- The number of branches equals the number of open-loop poles.
- The root locus plot is symmetrical about the real-axis.
- Real-axis segments are to the left of the odd numbered real-axis finite open-loop poles and zeroes.
- When there are more poles than zeroes, the extra branches will each approach an asymptote. These asymptote(s) have centroid $\sigma_a = \frac{\sum_{j=1}^n p_j - \sum_{i=1}^m z_i}{n - m}$ (i.e., the intersection point of all asymptotes on the real axis), and angles $\theta_a = \frac{(2k+1)\pi}{n - m}$, for $k = 0, 1, \dots, n - m - 1$.
- The real-axis break-in and break-away points can be found by setting $K^*(s) = \frac{-1}{L(s)}$, and solving $\frac{dK^*(s)}{ds} = 0$, for real values of s . Or, by writing the open-loop transfer function $K^*L(s) = K^* \frac{N(s)}{D(s)}$, and solving for the value(s) of s that satisfy $\frac{dD(s)}{ds}N(s) - D(s)\frac{dN(s)}{ds} = 0$.
- The imaginary-axis crossing (i.e., transition to instability) can be found by setting $K^*L(j\omega) = -1$, and solving for the values of ω that satisfy $\text{Re}\{L(s)\} = -1$ and $\text{Im}\{L(s)\} = 0$.

Table 1: Transforms of Functions

Time Functions	Transformed Functions
$\delta(t)$	1
$U(t)$	$\frac{1}{s}$
A	$\frac{A}{s}$
t	$\frac{1}{s^2}$
t^2	$\frac{2!}{s^3}$
t^n for $n = 1, 2, 3, \dots$	$\frac{n!}{s^{n+1}}$
e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$
t^2e^{-at}	$\frac{2!}{(s+a)^3}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{-at} \sin(\omega t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$
$e^{-at} \cos(\omega t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$
$e^{-at} \left[B \cos(\omega t) + \left(\frac{C - aB}{\omega} \right) \sin(\omega t) \right]$	$\frac{Bs + C}{(s+a)^2 + \omega^2}$
$2Ke^{-at} \cos(\omega t + \phi)$	$\frac{Ke^{j\phi}}{s+a-j\omega} + \frac{Ke^{-j\phi}}{s+a+j\omega}$

Table 2: Transform Properties

Time functions	Transformed functions
$f(t)$	$F(s)$
$af(t)$	$aF(s)$
$f(t) + g(t)$	$F(s) + G(s)$
$e^{-at}f(t)$	$F(s + a)$
$tf(t)$	$-\frac{d}{ds}F(s)$
$f(t/a)$	$aF(as)$
$[f(t - a)]U(t - a)$	$e^{-sa}F(s)$
$\dot{f}(t)$	$sF(s) - f(0)$
$\ddot{f}(t)$	$s^2F(s) - sf(0) - \dot{f}(0)$
$\dddot{f}(t)$	$s^3F(s) - s^2f(0) - s\dot{f}(0) - \ddot{f}(0)$
$\frac{d^n f}{dt^n}$ for $n = 1, 2, 3, \dots$	$s^n F(s) - s^{n-1}f(0) - \dots - sf^{(n-2)}(0) - f^{(n-1)}(0)$
$\int_0^t f(\lambda)d\lambda$	$\frac{1}{s}F(s)$
$f(0^+)$	$\lim_{s \rightarrow \infty} sF(s)$
$f(\infty)$	$\lim_{s \rightarrow 0} sF(s)$